Midpoints and segment bisectors

1 No

2 It is the midpoint of $\overline{BD}$.

3 Point K

4
a Point G
b $\overline{EF}$

5 $PR = 5.1 \text{ mm}$

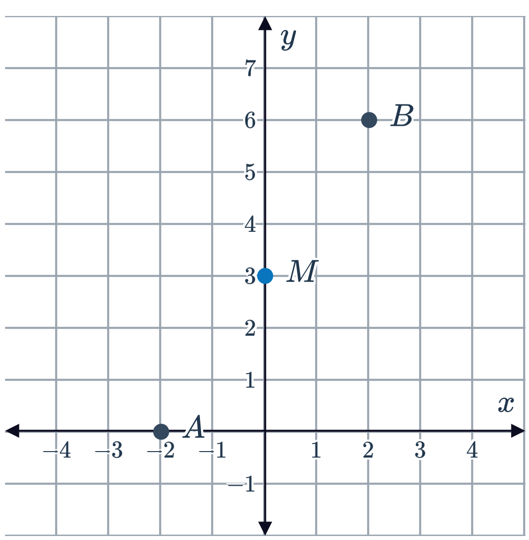
6 $AB = 51.4 \text{ cm}$

7
a $x = 9$
b $ML = 51$

8
a $k = 10$
b $XZ = 96$

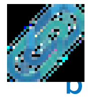
9 $\left(\dfrac{x_2 + x_1}{2}, \dfrac{y_2 + y_1}{2}\right)$

10
a 0
b 3
c



11

a $\left(\frac{1}{2}, \frac{5}{2}\right)$



b $(-11, -8)$

c $\left(\frac{5}{3}, \frac{1}{10}\right)$

d $\left(\frac{7}{2}, -\frac{7}{2}\right)$

12

a $(17, 10)$

b $(-3, -17)$

c $(9, -6)$

d $(11, -12)$

13 $(-6m, -4n)$

14 $(25^\circ \mathrm{N}, 97^\circ \mathrm{E})$

15 $\$28$ million

16 58 years

17

a $293\,319$

b $374\,632$

18 $PQ = 25$

19 $UV = 16x + 10$

20 $CD = 4\frac{2}{5}$ inches

21

a $\left(-\frac{15}{2}, 1\right)$

b $(-9, 6)$

c $\left(-\frac{21}{2}, 11\right)$

22

a $-2m - 4$

b $\frac{19n}{18}$

23

24

a $p=-\dfrac{3}{2}$

b $q=0$

Additional questions

25 $M=\left(\dfrac{x_1+x_2}{2},\dfrac{y_1+y_2}{2}\right)$

26

a $M\,(5,-2)$

b $M\,(10,6)$

27

a $M\,(5,7)$

b $M\left(-\dfrac{11}{2},-8\right)$

28

a $M\,(-3,2)$

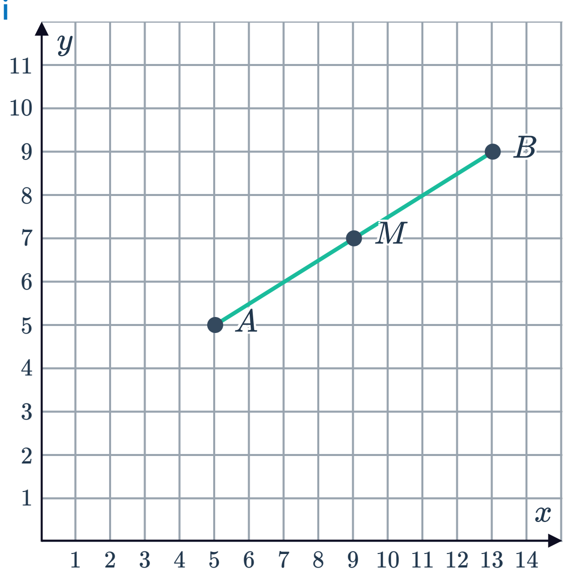
b $M\,(-3,-2)$

29

a

i $M\,(9,7)$

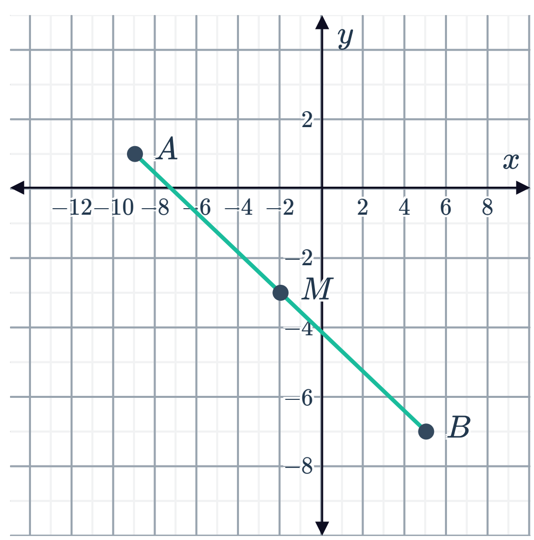
ii



b

i $M\,(-2,-3)$

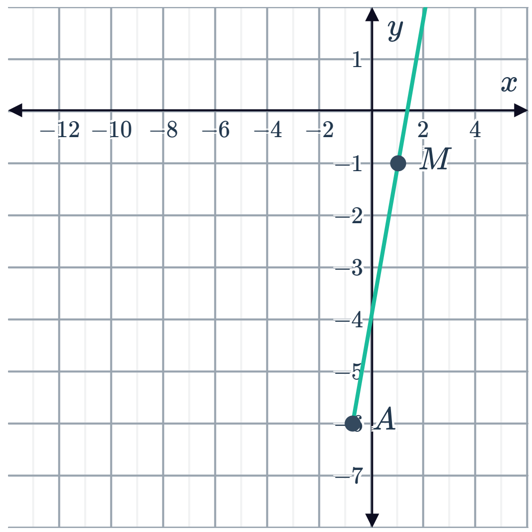
ii



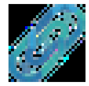
c

i $M\,(1,-1)$

ii

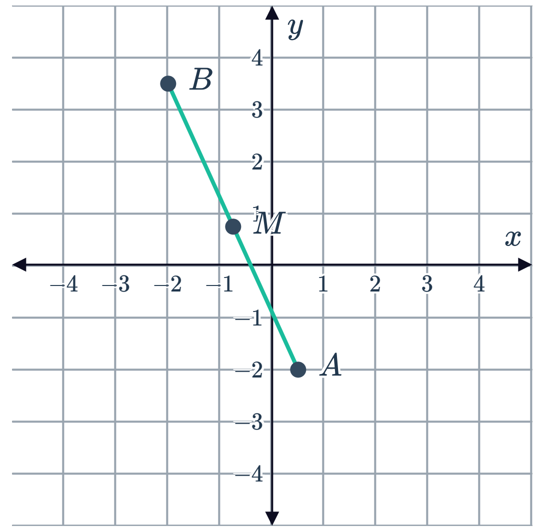


B



i $M\left(-\dfrac{3}{4},\dfrac{3}{4}\right)$

ii



30 $(4m,3n)$

31

a $A\,(5,6)$

b $A\,(4,-3)$

c $A\,(-12,-8)$

d $A\,(-4,-5)$

32 $(28^\circ\text{N},83^\circ\text{E})$

33 $Q\,(-6,1)\,,R\,(-8,4)\,,S\,(-10,7)$

34

a $p=-\dfrac{1}{4}$

b $q=0$

35

a $\overline{KL}\cong\overline{JK}$, $\overline{HL}\cong\overline{HI}$, and $\overline{EI}\cong\overline{IJ}$.

b $\overline{CD}$ bisects $\overline{JL}$ and $\overline{IL}$.
There are multiple equivalent names for $\overline{CD}$ and $\overline{GM}$.

c Point K is the midpoint of $\overline{LJ}$ and Point I is the midpoint of $\overline{EJ}$.

36 D is the midpoint of $\overline{CF}$, E is the midpoint of $\overline{DF}$ and $\overline{CB}$, and F is the midpoint of $\overline{DB}$.